Resolution Geometry: Undermining Spacetime

What of General Relativity's Spacetime Is Observed, What Is Selected, and What Repair Can Lock

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Abstract

Resolution Geometry (RG) cannot derive general relativity (GR): the source remainder leaves the Weyl curvature free, so the field equation is selected or measured. This paper finishes the converse programme. It does not derive spacetime; it undermines spacetime-as-an-object and states exactly what survives.

Undermine, repair, lock. An *undermine* is a pair of admissible spacetimes in one observational fibre that disagree on a spacetime claim. A *repair* is a new protocol that intersects the fibre. A *lock* is a fibre narrowed to a declared tolerance. The method rests on two imported batteries:

- *selection*: relaxation contracts the observational quotient but leaves null modes as initialized, so convergence is not identification (562/569 checks, failures retained);
- *lock automation*: adversarial witness search and active probe acquisition (892/892 after a disclosed repair; 240 acquisitions; 144 false point locks refuted).

The spacetime dispatch. Applied to a declared weak-field model of GR observables, the controller, which never sees the truth, returns:

- **LOCK**: spatial structure γ, via one Shapiro-class probe after box-type data left it in $[-2, 1.16]$; the nonlinearity β; a star's interior, via a helioseismic probe; and topology, via global light paths;
- **INCONSISTENT**: the Nordström class, which slow probes cannot reject;
- **permanent pool fibres**: black-hole interiors, metric values at coordinate points, and the Weyl curvature without a field equation.

Declaring the field equation turns tidal data into a Weyl lock.

Verdict. Spacetime is not refuted. It is *demoted*, from an observed object to a representative selected from an observational fibre by declarations. Its observable part locks with named protocols; its surplus is either repairable or permanently declared. Battery: 11 dispatch cases, 19 computed checks, 1 canary.

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1 Thesis and non-thesis

Non-thesis. GR's predictions are not challenged. Mercury, light deflection, Shapiro delay, GPS and Pound–Rebka all stand, and RG's own gravitation stack reproduces them *conditionally*. No rival dynamics is proposed. The physical path paper proved that RG cannot derive the field equation [1].

Thesis. The target is *spacetime as an object*: the claim that a complete Lorentzian manifold, with all its points, interiors, global shape and curvature, is what observation reveals. RG's claim is narrower and testable. Every part of that object is either:

- locked by a named protocol, or
- selected by a named declaration.

No third kind exists.

Definition 1.1 (Undermine, repair, lock)**.** Fix a declared model, a box (prior domain), calibration observations with error bounds, and a pool of admissible probes. A spacetime claim is a task h on the model's parameters.

(i) The claim is *undermined* when the feasible set F contains two states whose task values differ by more than 2δ: a witness pair.

(ii) A *repair* acquires a probe from the pool. The new constraint replaces F by $F' \subseteq F$.

(iii) The claim is *locked* when $\max_F h - \min_F h \le 2\delta$. It is *inconsistent* when $F = \varnothing$. It is a *pool fibre* when two states agree on every calibration and every available probe yet differ on h.

2 Machinery (imported, with retained failures)

Imported Theorem 2.1 (Selection is not identification; selection battery [4])**.** *Let $H = M^\top W M$ with W positive diagonal, and let $0 < \alpha < 2/\lambda_{\max}(H)$. The relaxation $x_{k+1} = x_k - \alpha(Hx_k - g)$ converges to $x^\dagger + P_{\ker M} x_0$. The observational quotient contracts, at rate $\rho = \max_{\lambda>0} |1 - \alpha\lambda|$, while the null component stays as it was initialized. A linear task $h^\top x$ is identified on an exact fibre iff $h^\top Z = 0$ for a basis Z of $\ker M$.*

The battery ran 1 152 trajectories, over 24 graphs with 16 starts each including ±100 offsets. It scored **562/569**. The seven failures are finite-budget non-convergence of slow-step cases, and their spectral iteration bounds (20 471 to 32 542 steps) exceed the 20 000-step cap; they are retained. An independent spectral replay reproduced all endpoints to 5.8×10^{-15}. The battery records its own milestones:

- a physical RG update law is **blocked**;
- GR observables are **not tested**;
- the attractor is inference, not physical time.

Imported Theorem 2.2 (Lock and repair; lock-automation battery [5])**.** *Let* $F = \{x \in box : |Mx-d| \le \varepsilon\}$, $L = \min_F h^\top x$ *and* $U = \max_F h^\top x$. *The midpoint has worst-case error* $(U-L)/2$, *so the task is locked to* δ *iff* $U - L \le 2\delta$. *A valid new observation intersects* F, *so* F *cannot grow; it may become empty if the model or error assumptions fail.*

The battery scored **892/892** over 192 cases with 240 acquisitions:

- 120 locks, with maximum locked error 3.3×10^{-6};
- 24 inconsistent cases, 24 budget-limited cases and 24 pool fibres;
- 144 naive point locks refuted.

The first implementation scored 870/892, failing by safe but wasteful dispatch; it is retained as v1. The repair was followed by regression testing, not a blind holdout. Primal/dual certificates were verified independently in floating point.

3 The undermine catalogue

	Spacetime claim	Undermine (witness)	Source
U1	Matter determines the field equation	Minkowski and Schwarzschild share $T = 0$; Nordström and Einstein vacua share $T = 0$	Phys. Path P12
U2	The freely falling box grounds GR	the box reads energy only: R_{0i0j} carries no γ, $R_{ijkl} \propto \gamma$; $0.876''$ (1911) vs $1.751''$ (1915)	Cornering F02
U3	Spacetime points are physical	the hole diffeomorphism moves values at points; the invariant relations are unchanged	Cornering F04
U4	The metric describes interiors and the global shape	Birkhoff: exteriors of equal mass agree; local data leave a global remainder	F05, L119
U5	Matter obeys the energy conditions	the conditions are forced only for kinetic sources; Λ, stiff matter and field tension lie outside the cone	Phys. Path P04
U6	G and M are geometric	geometry knows only GM/c^2	F07
U7	Distance is a spacetime property	radar, proper and optical distances differ	Light L09
U8	One potential describes gravity	Newton fails exactly where structure is resolved ($\varepsilon_{\text{crit}}$)	Duality
U9	The full curvature is observed	Weyl is the gravitational remainder, free given the sources	Phys. Path

4 The spacetime dispatch battery

Declared model. The model is linearized weak field, with parameter vector

$$x = (\gamma,\ \beta,\ c_\star,\ c_{\mathrm{BH}},\ \tau,\ g_{\mathrm{coord}},\ w,\ r,\ s)$$

and box $[-2, 2]^9$. The entries are:

- spatial structure and nonlinearity (γ, β);
- a star-interior moment and a black-hole-interior moment;
- an inverse topology scale;
- a coordinate metric value (gauge);
- the Weyl-electric part, the Ricci part and the source.

The synthetic truth is GR: $\gamma = \beta = 1$ and $r = \kappa s$ with $\kappa = 1$.

Protocols. Each protocol is a row $y = c_0 + a \cdot x \pm \varepsilon$.

- **Slow-probe deflection.** $1/v^2 + \gamma$ at $v/c = 0.01$ and 0.03, relative error 5×10^{-4} and 3×10^{-4}.
- **Clock redshift.** Independent of γ.
- **Light deflection and Shapiro delay.** Both scale as $1+\gamma$, with errors 10^{-3} and 2×10^{-5}.
- **Perihelion.** Scales as $(2 + 2\gamma - \beta)/3$.
- **Helioseismology.** Reaches a star's interior only.
- **Matched circles.** Global light paths reaching the topology.
- **Gradiometer.** Measures the tidal tensor, which is Weyl plus Ricci.
- **Source measurement.**
- **The field equation $r = \kappa s$.** This is a *declaration*, not a probe.

The controller reimplements the semantics of the lock-automation controller with tolerance $\delta = 10^{-3}$. It never sees the truth.

Case	Spacetime claim	Initial interval	Repair acquired	Final status
G1	the box fixes spatial structure γ	$[-2, 1.160]$	Shapiro	LOCK $1.00001 \pm 2 \times 10^{-5}$
G2	the metric data fix the nonlinearity β	$[-2, 2]$	perihelion, Shapiro	LOCK 1.0003 $\pm 3.4 \times 10^{-4}$
G3a	the exterior fixes a star's interior	$[-2, 2]$	— (no pool probe separates)	POOL FIBRE (gap 4.0)
G3b	... repaired by an interior protocol	$[-2, 2]$	helioseismology	LOCK 0.3000 ± 10^{-4}
G4	the exterior fixes a black-hole interior	$[-2, 2]$	— (no admissible probe)	POOL FIBRE (permanent)
G5a	local data fix global topology	$[-2, 2]$	—	POOL FIBRE

Case	Spacetime claim	Initial interval	Repair acquired	Final status
G5b	... repaired by global light paths	$[-2, 2]$	matched circles	LOCK 0.2001 ± 10^{-4}
G6	slow probes reject the Nordström class	$[-1, -0.98]$ (consistent)	Shapiro	INCONSISTENT
G7	a metric value at a coordinate point is observable	$[-2, 2]$	— (gauge)	POOL FIBRE (permanent)
G8a	tidal data fix the Weyl curvature	$[-1.25, 2]$	—	POOL FIBRE (gap 3.25)
G8b	... with the field equation *declared*	$[0.4999, 0.5003]$	—	LOCK $0.5001 \pm 2 \times 10^{-4}$

All five locks cover the truth. A canary that acquires every protocol still cannot lock the gauge value.

5 Theorems

Theorem 5.1 (Spacetime is a selected representative)**.** *In the declared class, consider a full spacetime used to describe a region. Its parameters split into two kinds:*

- *those fixed by the observational quotient of the available protocols, which lock (G1, G2, G3b, G5b);*
- *those fixed only by declarations: the field equation (G8), the gauge and coordinates (G7), interior or global data beyond every admissible probe (G4, G5a), and the box.*

By the selection theorem, any algorithm that produces the full spacetime from the data returns a representative whose null components are inherited from its declarations or its initialization, not from observation.

Proof. Write the parameter space as the quotient directions plus $\ker M$. Each quotient direction is either locked by an available probe or is in the pool's kernel. For kernel directions, the pool-fibre LP exhibits two admissible states with identical outputs on every calibration and probe but different task values (G3a, G4, G5a, G7, G8a). The selection theorem shows that relaxation leaves exactly those components as initialized. □

Theorem 5.2 (Repair is information, never deduction)**.** *Every lock in the battery is preceded either by an acquired probe (G1, G2, G3b, G5b) or by a declaration (G8b). Each new constraint gives $F' \subseteq F$; no amount of computation on fixed data narrows F. Declaring a field equation* does *narrow F, but it is a declaration, and RG types it as such: it is not an observation.*

Theorem 5.3 (Permanent fibres)**.** *In the declared class, three spacetime claims have no admissible repair:*

- *metric values at coordinate points (G7): diffeomorphism gauge, since no invariant observable depends on them;*
- *black-hole interiors for exterior observers (G4): no causal probe returns across the horizon;*
- *global topology beyond the observable horizon (G5a, when global light paths are unavailable).*

Their status is declaration*, permanently.*

Theorem 5.4 (The field equation converts tidal data into Weyl data)**.** *A gradiometer measures R_{0i0j}, which is the Weyl-electric part plus a Ricci part. Without a declared field equation the Weyl part lies in a fibre of width* 3.25 *in the box (G8a). Declaring $r = \kappa s$ locks it (G8b). Vacuum is the special case $s = 0$, where "tidal = Weyl" holds only because Ricci-flatness is declared. This is the Weyl remainder (U9) in operational form: it is reachable by tidal protocols* plus *a field equation, never by source data alone.*

Theorem 5.5 (Rival theories are rejected by structure probes)**.** *The Nordström class ($\gamma \approx -1$) stays consistent with slow-probe deflections and clocks (G6, initial feasible set nonempty). A single Shapiro-class probe makes it inconsistent. This is the duality theorem in dispatch form [2]: box-type data read the energy footprint, and structure-sensitive data select among field equations.*

6 What survives: the spacetime ledger

Component of spacetime	Status	Protocol that locks it	Declaration needed
Light cones and conformal class	locked (cone stratum)	null data, L64–L68	H-top, H-int
Lapse N (clock rates, redshift)	locked	clocks, Pound–Rebka	time normalization
Spatial structure h (γ)	locked	light, Shapiro, two-speed panels	—
Nonlinearity (β)	locked	perihelion with γ	—
Weyl-electric curvature	conditional lock	gradiometer	field equation
Star interior	repairable	helioseismology, neutrinos	—
Global topology (within the horizon)	repairable	matched circles	—
Field equation	selected	rival rejection by structure probes	Lovelock or data
Energy conditions	declared (for non-kinetic sources)	kinetic cone for particle sources	source class
Metric values at points	permanent declaration	none (gauge)	coordinates
Black-hole interior	permanent declaration	none for exterior observers	interior model
Topology beyond the horizon	permanent declaration	none	topology
Units (G, M separately)	declaration	counted channel plus SI	unit anchor

Verdict 6.1 (Spacetime). Spacetime is not refuted. It is *demoted* from observed object to selected representative.

- What locks is *observable spacetime*: light cones, clock rates, spatial structure, nonlinearity, tidal curvature given a field equation, and deflections and delays.
- What is undermined is *spacetime realism's surplus*: substantival points, interiors behind horizons, topology beyond reach, Weyl data without protocols, and the field equation presented as derived.

- Every item of the surplus is either repairable by a named protocol or permanently a declaration. No third kind exists in the declared class.

7 Blocked milestones (stated, not hidden)

1. **A physical RG update law: blocked.** The selection attractor is a declared quadratic inference algorithm, not a law of nature, and its iteration is not physical time.
2. **GR built from attractors instead of spacetime: not shown.** No GR observable is recovered from attractors.
3. **The source-remainder no-go stands.** RG still cannot derive the field equation. This paper undermines the ontology, not the physics.
4. **Linear weak-field scope.** The dispatch model is declared and linearized. Strong-field, stationary (gravitomagnetic, the Weyl-magnetic part) and dynamical spacetimes are not dispatched.

8 Ladder of the undermine path

Rung	Statement	Status	Witness
$\Upsilon 0$	Declarations: model, box, protocol pool, tolerance	declared	—
$\Upsilon 1$	Selection contracts the quotient; null modes persist; convergence $\neq$ identification	I (selection)	562/569
$\Upsilon 2$	Lock iff width $\leq 2\delta$; repair never enlarges F	I (lock)	892/892
$\Upsilon 3$	Undermine catalogue U1–U9	imported	prior papers
$\Upsilon 4$	Structure γ locks only with structure probes	F	G1
$\Upsilon 5$	β locks with perihelion and structure	F	G2
$\Upsilon 6$	Star interior: pool fibre, repaired by an interior probe	F	G3a, G3b
$\Upsilon 7$	Black-hole interior: permanent fibre	N	G4
$\Upsilon 8$	Topology: pool fibre, repaired by global light paths	F	G5a, G5b
$\Upsilon 9$	Rival class rejected only by structure probes	F	G6
$\Upsilon 10$	Gauge values: permanent fibre	N	G7, canary
$\Upsilon 11$	The field equation converts tidal data into Weyl	H: field eq.	G8a, G8b
$\Upsilon 12$	Spacetime is a selected representative	F (declared class)	Thm 5.1
$\Upsilon 13$	Physical RG update law; GR from attractors; strong and dynamical fields	O / blocked	—

9 Battery and limits

`rg_spacetime_undermine.py` (NumPy, SciPy; seed 20261028) runs the 11 dispatch cases, 19 computed checks and one canary, which requires that the gauge never locks. Two batteries are imported unchanged, with their packages, retained failures and verifiers:

- `rg_selection_battery.py` (562/569);
- `rg_lock_automation.py` (892/892; v1 870/892).

Limits. All acquisitions are synthetic. The weak-field coefficients are standard parametrized post-Newtonian (PPN) scalings, used as a declared linear model, not fitted to data. The box is a substantive prior. Floating-point linear programs are not interval-certified. The battery is not blind.

10 Non-claims

1. GR is not refuted; no rival dynamics is proposed.
2. The paper does not claim that spacetime does not exist; it claims that spacetime's surplus over the observable quotient is declared.
3. No empirical dataset is analysed; Cassini-class errors are illustrative.
4. Novelty is claimed only for the dispatch typing of spacetime claims and its assembly with RG's batteries.

References

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